

Chapter 11

Monitoring of Chemicals and Water

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11.1 CHEMICAL AND WATER MONITORING OVERVIEW

The relationship between the water and chemical purity with respect to high-yielding wafers is well known in the semiconductor industry. Chemical and water monitoring can improve process control and prevent excursions that lead to scrap and reduced yield [1]. Thus, monitoring these fluids is critical for identifying contamination and keeping the chemical concentration of processing baths stable to optimize the contamination removal. In the early era of semiconductor manufacturing, temperature and time were monitored during the wet processing steps, whereas concentration and chemical purity were hoped to be adequate for the cleaning, etching, and surface conditioning tasks at hand.

Now, the integrated circuit (IC) industry realizes the relationship between chemical concentration and its ability to remove contamination. In addition, monitoring ultrapure water (UPW), not only to confirm its ultrapure state but also to assure that the chemicals used for wet processing steps are completely removed during the rinsing process, is critical for contamination-free manufacturing. In this chapter, concentration monitoring will be discussed along with water purity monitoring. In-line, also called on-line, water and chemical monitoring will be the focus, where the monitoring system is integrated onto the processing tool.

The chemical and water monitoring techniques include (1) titration, (2) refractometry, (3) conductivity and resistivity, and (4) optical absorption spectroscopy.

Other techniques exist for monitoring chemicals and concentration but will not be discussed in this chapter. Methods include acoustics (i.e., ultrasonic) for measuring chemical density [2] for solutions, such as concentrated H₂SO₄ [3], and ion-selective electrodes, such as pH electrodes or fluoride electrodes [4].

The equipment used to monitor the water and chemicals must give accurate quantitative results, have reasonable cost, and have high reliability. The in-line monitors must be noninvasive and integrate onto the wet chemical processor, recirculation loop, or delivery line with ease. The ability to use the monitor at the point of use for a chemical bath is important. The software must seamlessly interface the monitoring system with the chemical delivery or wet

processing tool. The time response must be faster than expected process changes, usually in the range of a few seconds or minutes for “real-time” monitoring, and must provide a continuous data stream. Importantly, the monitoring technique must be compatible with the chemical being monitored. The sensor must have high resolution and accurately measure a wide range of concentrations. Plus, the monitor must not be a source of contamination itself.

11.2 CHEMICAL MONITORING

11.2.1 Chemical Monitoring Techniques

An engineer can monitor chemical concentration by grabbing samples from the process tool and taking them to a laboratory for analysis. This entails some work, carries a risk of exposure to hazardous chemicals, and most significantly, requires considerable time before results are reported. This time delay can be critical, as many wafers may be processed while waiting for the laboratory results. In-line chemical monitors give continuous chemical concentration readings, with a time interval in the range of seconds to minutes, because of the fast response of the measurement tool. Thus in-line methods of chemical analysis for monitoring and measurements have been developed and will be discussed in this chapter.

There are many methods of measuring chemical concentrations, in which one subset is in-line chemical monitoring. These methods are discussed in this section.

Titration can be used to quantitatively monitor a variety of chemicals, such as aqueous solution, which include acids, bases, and oxidizing agents. Autotitrators, as the name implies, automatically remove a small sample, in which the analyte is dissolved, from the process tank, then titrate the sample with reagents, termed the titrant: HCl for NH_4OH (ammonium hydroxide) in SC-1 (standard clean 1), KMnO_4 (potassium permanganate) for H_2O_2 (hydrogen peroxide), and Karl Fisher reagent for water content [5]. The sensors used for titration must be appropriate for the analysis technique, for example, a pH electrode, a potentiometer, or a color sensor. Autotitration is a high-maintenance monitoring method, requiring periodic “feeding” of the measurement device with chemicals and regular replacement of electrodes. For these reasons, it is rarely used in fabs today. However, the result of the titration yields a precise value for the analysis being performed.

Refractometry is used to measure the index of refraction (RI) of a fluid, which in turn can be used to obtain information about its composition, concentration, and purity. The RI is an optical property of materials, which affects, among other things, to what degree the chosen wavelength of light bends when passing through a substance, usually measured by determining the critical angle of refraction using a refractometer [6]. It is most useful in a binary mixture, as each component in a mix can contribute to the overall refractive

index. Some materials that do not ionize in an aqueous solution can be conveniently measured through their refractive index, for example, H_2O_2 [7]. Monitoring concentration by means of refractive index measurement is temperature sensitive. The measure of the RI of chemicals and solutions for etching, cleaning, and surface conditioning, along with chemical-mechanical polishing slurry and electrochemical deposition solution, is used in semiconductor manufacturing. Commercially available components can be integrated onto the processing tool or delivery system.

Conductivity and **resistivity** are the measures of the electric current, as it passes through a solution. Ions are needed to conduct electricity through an aqueous solution. Therefore conductivity measures the amount of active ions, the electrolytes, in a solution, that can arise from salts, acids, and bases, which dissociate to form ions [8]. Conductivity measurements cannot distinguish between the compositions of the ions in solution, only that there are ions present. In addition, the processing by-products can contribute to the conductivity of the solution, not just the processing reagents. As with most monitoring techniques, temperature affects the measurement. Conductivity can be used for weak and strong electrolytes, and because conductivity increases with temperature, more ions are in solution at increased temperatures. Components for conductivity measurement of wet processing solutions are commercially available.

pH is the measure of the acidity or alkalinity of an aqueous solution, which is based on the number of hydrogen ions (H^+) or hydroxyl ions (OH^-) in the solution [9]. In water, acids dissociate to produce H^+ , whereas bases dissociate to produce OH^- . The amount of these ions in the solution is related to the pH:

$$\text{pH} = -\log_{10}[\text{H}^+] \quad (11.2-1)$$

where $[\text{H}^+]$ is the concentration of hydrogen ions. The range for pH is 0–14, with $\text{pH} = 7$ being neutral. A low pH, <7 , indicates the solution is acidic; a high pH, >7 , indicates the solution is alkaline. Electrodes for measuring pH of aqueous process solutions are commercially available; however, glass electrodes are usually used and make it difficult to measure HF (hydrofluoric acid) solutions, although HF-specific electrodes are available. Again, the pH is temperature dependent.

Optical spectroscopy is based on the absorption of electromagnetic radiation, usually in the ultraviolet (UV), visible, or infrared (IR) portion of the spectra by a material. In particular, near-infrared (NIR) spectroscopy is based on the absorption of electromagnetic radiation at wavelengths in the range of 800–2500 nm [10,11]. The vibration of covalent bonds within a molecule is especially pronounced in the NIR and IR, which include C–H, O–H, and N–H. These wavelengths can be used to monitor chemical solutions. Furthermore, acids and bases interact with water molecule and thus influence the chemical bonds of water, allowing NIR to also monitor these solutions

[12,13]. With respect to optical absorption, this chapter will focus mainly on NIR.

11.2.2 Wet Chemical Monitoring

Chemical and water monitoring can be integrated into wet processing equipment; wet bench batch immersion processing tanks, single wafer process modules, rinsing modules, and incoming chemicals are all amenable to various methods of monitoring. The motivations for in-line chemical concentration monitoring for wet processes include (1) improved process control, (2) savings on chemicals and chemical disposal by increasing bath life, and (3) increased tool uptime.

If used in a closed-loop control system, such as a wet bench tank, monitoring helps maintain the ideal concentrations of the process chemical or chemicals. For example, this is especially important for sulfuric acid (H_2SO_4), H_2O_2 mixture (SPM), and SC-1 (APM—ammonium hydroxide + H_2O_2 mixture), where the chemicals have high vapor pressures, such as $\text{NH}_{3(\text{aq})}$ (aqueous ammonia, NH_4OH), or decompose, such as H_2O_2 , which must be replenished over time.

In closed-loop control systems, feedback from the process monitor is used to take corrective action and maintain constant process parameters. In the case of chemical monitoring in wet processes, the output of the monitoring instrument is used by the tool controller to create a stable process. Monitoring is used to extend the bath life, by determining the decrease (or, in some cases, increase) in concentration over time during the lifetime of the bath and, sometimes during the processing step, by the replenishment of the chemical. A bleed and feed method is employed to dispense a chemical solution into the active bath with the goal of achieving a constant concentration of the reagents and reaction products over a long period. For example, since H_2O_2 in SC-1 deteriorates over time, it is necessary to periodically add it to the bath. The tool controller reads the current concentration of H_2O_2 from the monitor and if it is below an acceptable value, the controller operates a valve to add a small amount of H_2O_2 to the bath, termed spiking.

The use of chemical monitoring often proves useful when a new process is being introduced into the production line where the optimum process parameter ranges must be determined. Chemical concentrations can be correlated to the results, such as the etching rate and even device performance, as part of understanding a new process's behavior.

Chemical costs, and often chemical disposal costs, can be significant [14]. This is especially true for organic photoresist strippers, which are generally proprietary mixtures, and require special hazardous waste disposal. Accurate and continuous chemical monitoring can enable longer bath life, thus saving on both chemical and chemical disposal costs as well as increasing the tool utilization through increased bath life [15].

Process excursions are sometimes caused by component failure in wet process tools. Common examples are malfunctioning valves or failing pumps. These excursions can cause expensive damage to wafers, damage that may not be discovered for some time. In extreme cases there can be serious financial implications to such an event. Chemical monitoring, even with a simple alarm, can notify engineers that there is a problem long before extensive damage is caused.

11.2.3 Chemical Monitoring Procedure

There are a variety of wet chemicals, mixtures, and proprietary formulations used in semiconductor cleaning, surface conditioning, and etching processes. In addition, water is ubiquitously monitored for purity. Typical wet processing chemicals used in semiconductor manufacturing and discussed in this chapter are summarized in [Table 11.2-1](#) and outlined in the following discussion:

SC-1 is a common cleaning process, outlined in Chapter 4. The chemicals, NH_4OH and H_2O_2 , are used at various concentrations and

TABLE 11.2-1 Chemicals, Solutions, and Formulations	
Process	Chemicals to Monitor
SC-1	H_2O_2 NH_4OH
SC-2	H_2O_2 HCl
HF/BOE (buffered oxide etch)	HF NH_4F
SPM	H_2SO_4 H_2O_2
DSP	H_2SO_4 H_2O_2
DSP+	H_2SO_4 H_2O_2 HF
Aluminum-compatible photoresist strippers/residue removers	H_2O NH_2OH (hydroxylamine) NMP (n-methylpyrrolidone) DMSO Proprietary formulations
Copper-compatible photoresist strippers/residue removers	H_2O_2 Proprietary formulations
DSP, dilute sulfuric peroxide; SC, standard clean; SPM, sulfuric acid and hydrogen peroxide mixture.	

temperatures. Both components are unstable, especially at the high processing temperatures needed for this solution, for example, $>65^{\circ}\text{C}$. H_2O_2 decomposes into water (H_2O) and oxygen (O_2), especially in a basic solution such as SC-1, and ammonia evaporates over time. As a result, SC-1 cannot be purchased as a ready-made mixture, it must be blended on-site. Chemical monitoring can verify that the mixture has the proper proportions. The concentrations of the SC-1 components can slowly drop over time, and will do so faster during the cleaning process, especially in a hot process bath or a high chemical concentration bath. Chemical monitoring has the ability to keep track of these changes and trigger spiking of additional chemicals as needed [16].

SC-2 is a mixture of hydrochloric acid (HCl) and H_2O_2 , also known as HPM (a mixture of hydrochloric acid and hydrogen peroxide) or dilute HCl , which are widely used and are mixed on-site, so chemical monitoring is appropriate for it as well. H_2O_2 is more stable in an acidic environment than in a basic one, as explained in Chapter 4. In addition, although HCl is not as volatile as NH_3 , it still has a significant vapor pressure, at temperatures that exceed 50°C . Although spiking of HCl or H_2O_2 into SC-2 or dilute HCl is not as common as with the spiking into SC-1, monitoring the bath to assure the reagents are at their optimum concentration is critical for adequate metal contamination removal.

Hydrofluoric acid (HF) and buffered hydrofluoric acid (BHF— NH_4F + HF) solutions are regularly used in semiconductor device manufacturing, both for surface conditioning and dielectric etching. The vapor pressure of HF is low, so little evaporation occurs; however, a significant etch product buildup that causes a decrease in etching rate can occur. For processes that combine other chemicals with HF, for example, HF/ HCl , HF/ H_2O_2 , or as discussed later in this section, $\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2/\text{HF}$, special monitoring is required.

Sulfuric acid and hydrogen peroxide mixture (SPM), sometimes called Piranha, is another unstable mixture that is prepared on site, typically for photoresist stripping and residue removal. Fresh mixing of H_2SO_4 and H_2O_2 solutions spontaneously heat to boiling, and the process generally takes place at about 120°C . The amount of H_2O_2 to that of H_2SO_4 in the mixture is about 1:2 to 1:8. These components react to form Caro's acid. Hydrogen peroxide is typically added before or during wafer processing. Monitoring can both verify that the mix is usable and help manage H_2O_2 spiking volumes.

Dilute sulfuric peroxide (DSP) is not as unstable as the more concentrated SPM solution. It is used for back-end-of-line (BEOL) postetch aluminum (Al) stripping. DSP+ is DSP with ppm (part per million) levels of HF added. Chapter 4 covers the DSP and DSP+ mixtures. The HF concentration is critical for this process.

HF in DSP+ cannot be measured spectroscopically. Special methods have been developed to monitor this process.

Photoresist strippers and residue removers include simple chemicals, such as NMP and DMSO, and proprietary blends of chemicals. Many of these include H_2O , which must be within a specific concentration range to avoid corrosion of metal. Some Al postetch residue removers include hydroxylamine (NH_2OH) as an active ingredient. Both H_2O and NH_2OH concentrations can be monitored. In some cases, with proper calibration, other components in the mix may be monitored as well. Copper-compatible strippers include a variety of formulations, each of which has its own monitoring requirements. These vary depending on the formulation and can include H_2O , H_2O_2 , various organic components, and rarely HF.

Ultrapure water (UPW), in the case of semiconductor processing deionized water (DIW) monitoring using resistivity, is commonplace. The water is monitored not only at the point of use but also upon production, upon arrival at the processing tool, and while wafers are processed. Any trace contamination in the water will cause the resistivity to drop; thus resistivity is a good judge of the purity. As with most measurements, temperature affects the value; higher temperatures increase the conductivity, which is equivalent to a decrease in resistivity.

11.3 OPTICAL ABSORPTION SPECTROSCOPY

11.3.1 Optical Absorption Spectroscopy Overview

Absorption spectroscopy measures how much light is absorbed by a sample over a range of wavelengths defined by the electromagnetic spectra [17]. The wavelength at which a chemical component absorbs light is a function of the type of ion or molecule that is being monitored. The amount of light absorbed is related to both the type and concentration of the chemical through the Beer–Lambert law [18]. There are several types of interactions that can occur between light and chemicals, in which each type typically occurs within a particular wavelength range. The proportion of light that passes through a sample is called the transmission through the sample, whereas the absorbance is the negative log of the transmission, and is linearly dependent on the concentration of the chemical discussed in [Section 11.3.3](#).

The well-known laws of classical physics describe the motion of large objects, such as billiard balls, but do not properly describe the behavior of atoms and molecules. On this microscopic level the laws of quantum mechanics apply. The atoms and molecules do not have a continuous range of available energy states and thus can only absorb photons of specific energies, quantized state, that enable transition between these energy states [19].

For a molecule made of two atoms, such as HF, the two atoms will vibrate relative to each other, similar to two balls connected by a spring. Energy, in the form of light, will cause this molecule to vibrate. However, a quantum system cannot vibrate at every possible energy level. It can vibrate only with particular

energies, thus particular wavelengths of light. Another way of expressing this is that the molecule can occupy only particular vibrational energy levels. As a result, the molecule will absorb light only at the exact wavelength that has the right amount of energy to bring it to a higher energy level [20].

Energy is related to the wavelength of light through Planck's equation [21]:

$$E = h\nu \quad (11.3-2)$$

where E is the energy, h is Planck's constant, and ν is the frequency, which is related to the wavelength by the following equation:

$$\nu = c/\lambda \quad (11.3-3)$$

where c is the speed of light in a vacuum and λ is the wavelength.

When a molecule absorbs light, this absorption can be plotted with respect to its spectral energies. These spectra can be divided into energy ranges, depending on what molecular or atomic interactions are involved: the different interactions give rise to different energy transitions and therefore different ranges of photon energies correlated to the wavelengths of absorption [22]. Transitions considered to be in the optical wavelengths are related primarily to midenergy photons. Photons that are out of this optical range include extremely high-energy photons, which can displace inner orbital electrons in atoms. These interactions are useful for X-ray analysis [23]. The lowest energy photons change the rotational state of molecules, which in some applications, such as microwave spectroscopy, can be a useful method of analysis [24].

In the optical range, high-energy (short wavelength) interactions are due to electronic transitions. In these transitions an electron is moved from a low-energy orbital to a higher energy one. These appear in the UV or visible regions of the energy spectrum. This is a common method of analysis in laboratories, which can be useful for in-line monitoring of some metals in aqueous solutions [25]. Absorption in the UV can also be used to monitor hydrogen peroxide [26,27].

When a photon in the low-energy IR range is absorbed by a molecule, it can change the degree of vibration, also called the "vibrational state" of a molecule. The precise wavelength required to do this depends on the molecule's structure and physical properties, for example, the weight of its atoms, the strength of the chemical bonds between them, and the molecule's interaction with its surroundings. This dependence on a molecule's structure and environment means that different molecules have specific patterns of IR absorbance. The pattern typically includes regions of high absorption that are termed absorption peaks. In many cases the pattern is quite specific to the molecule, so that the IR region includes a "fingerprint region" typically defined at wavenumbers in the range of $1500\text{--}650\text{ cm}^{-1}$ ($\sim 6700\text{--}15,000\text{ nm}$) [28]. This fingerprint region is typically in the far IR that for technical reasons, such as the availability of transparent materials in this range, is impractical for in-line analysis. However, it is often used in laboratory analysis of organic materials [29].

11.3.2 Molecular Vibrations and Wavelength Ranges

The vibrations in the far IR are the fundamental vibrations of a molecule, the lowest energy vibrational transitions that it can undergo. However, molecules can undergo transitions to higher energy vibrational states as well; these transitions are at higher energies, which correspond to absorption of photons of higher energy and shorter wavelengths in the near-IR region (750–2500 nm). These absorbances are termed overtones. This range of wavelengths has technical advantages: readily available optical materials, such as quartz and sapphire, which are transparent at these wavelengths; affordable low-noise photodetectors; and fiber optics are available for use [30].

Absorption peaks of overtones are broader, and can overlap, and therefore special mathematical techniques are needed to extract the relevant information from the spectra.

There is one more important point concerning ions. While molecules have chemical bonds, and thus each molecule has clearly defined molecular vibrations, ions that have chemical bonds, such as NH_4^+ , will also have vibrations. Weak acids, such as HF, do not dissociate completely in an aqueous solution, and will have molecular vibrations as well. However, simple inorganic ionic chemicals, such as HCl, can be measured using NIR, despite the fact that it is a strong acid and completely ionizes in H_2O [30] to H^+ and Cl^- ions, and there is no “molecule” to speak of, or molecular bonds that can vibrate.

An ion in a water solution interacts with the water molecules, so the water molecules’ vibrations are affected by the presence of the acid ions. In addition, H_2O itself has a strong signal at some wavelengths in the NIR [31], shown in Fig. 11.3-1. Thus, lack of water can be easily detected, that is, 99 wt% (weight %) looks different than 100 wt% H_2O spectra of H_2O , irrespective of what the other 1 wt% is, or whether it has a signal in the NIR [31].

11.3.3 Beer–Lambert Law

There is a simple mathematical relationship between the concentration of a chemical and its absorbance at any single wavelength, the Beer–Lambert law [18]:

$$I_l = I_0 e^{-\alpha c l} \quad (11.3-4)$$

where I_l is the intensity of light coming out of the flow cell, I_0 is the intensity of light going into the flow cell, α is the specific absorption of the material in $\text{L mol}^{-1} \text{cm}^{-1}$, c is the concentration of material in mol L^{-1} , and l is the length of the flow cell in cm. For historical reasons, the log base 10 is sometimes used instead of the natural log [32]. Note that the Beer–Lambert law shows a linear relationship between the log of the transmission and the concentration:

$$c = \frac{\log\left(\frac{I_1}{I_0}\right)}{\alpha l} \quad (11.3-5)$$

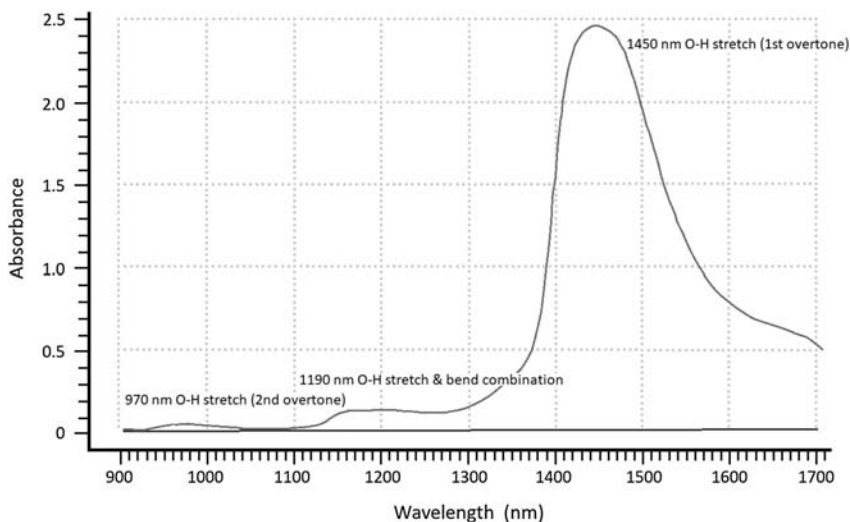


FIGURE 11.3-1 Water absorption in the near-infrared region of the spectra, showing the three distinct absorption wavelengths. Adapted from E. Shalyt, G. Liang, P. Bratin, C. Lin, in: M.K. Balazs (Ed.), *Semiconductor Pure Water and Chemical Conference Proceedings (SPWCC)*, 2007, p. 19 by the authors.

The Beer–Lambert law holds for mixtures of chemicals, as long as there are not chemical reactions or physical interactions between the various chemical species. Thus, like most chemical “laws,” it works best at low concentrations. This is the basis for optical chemical analysis.

For simple mixtures, with only one component, it is often possible to find a single wavelength whose absorbance is linear with the concentration. Fig. 11.3-2 shows the concentration, in weight percent (wt%), of H_2O in a chemical mixture versus absorbance, depicting the linear relationship of the Beer–Lambert law.

11.3.4 Complex Mixtures

Chemometrics is defined as the ability to derive data from chemical system. In the case of optical absorption, chemometrics is based on the assumption that the Beer–Lambert law holds, that is, the absorption of light at any wavelength is linearly proportional to the concentration of the chemical. In addition, chemometrics assumes that in a mixture of chemicals, the absorption at any wavelength is a linear combination of the absorbances due to each chemical component. Although this is generally true, there are exceptions. In complex mixtures, absorption peaks overlap, although the Beer–Lambert law does apply, it is not practical to use it, as there is no single wavelength that can be

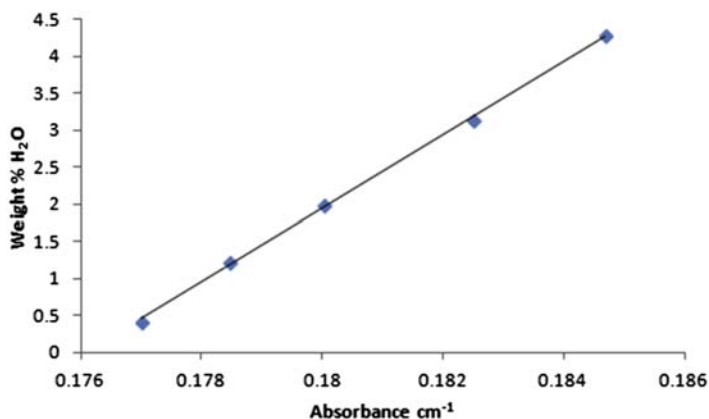


FIGURE 11.3-2 Graphical representation of the concentration, in weight percent (wt%), of H_2O in a chemical mixture versus absorption, showing the linear relationship of the Beer–Lambert law. *Used with permission of the authors.*

used for the calibration. Thus chemometrics, a method of multivariate analysis, is used to correlate the spectrum with the concentration [33]. By using this method, information from the entire spectrum, not just one wavelength, can be used to extract the relevant information, such as the concentrations of the chemicals in the solution. Fig. 11.3-3 shows an example of a series of complex spectra that includes information about several components that can be analyzed using chemometrics.

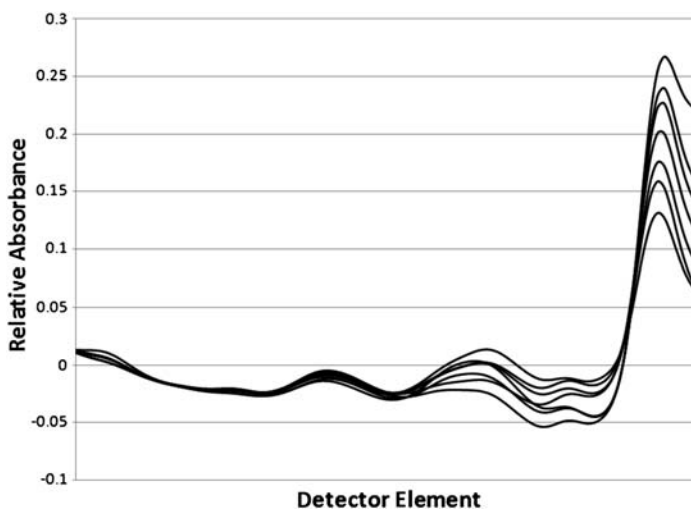


FIGURE 11.3-3 A series of complex absorption spectra that includes information about several components that can be analyzed using chemometrics. The detector element is the detector array. *Used with permission of the authors.*

11.3.5 Calibration

11.3.5.1 Calibration Model

The most common way to calibrate an optical absorption system for chemical concentration is to use the partial least squares (PLS) fit algorithm. The PLS algorithm chooses from a family of curves, the curve closest to values both set and empirically observed for a particular problem. This in turn is a special case of a mathematical method called principal component analysis (PCA). This system is defined by a set of equations. PCA enables modeling and identification of the most important variables associated with these equations that are used to define. PCA rates the importance, or weight, of the variables in the system. The importance of a variable is to what degree it changes the system; this is the weighing factor; a variable that changes the output to a large degree has a higher weighing factor than a variable that does not significantly change the output. In this case of PCA, the chemical system is the process formulation, such as SC-1, in which the concentrations of the components are the variables that are analyzed by optical absorption, which is the output, at various wavelengths [34].

PCA is a means of analyzing data and reducing the number of dimensions that must be dealt with. The PLS fit goes a step further and correlates the data with a matching set of reference data. The PLS method is commonly used in chemometrics to correlate the measured spectra with the reference concentration data. The net result of this correlation is a calibration model, a vector that is weighted for the different wavelengths of the spectrum [35].

An example of PCA applied to a set of chemical samples, is as follows: each chemical sample has an NIR spectrum of 100 points; thus each spectrum can be described as a point in a 100-dimensional space, each dimension representing a wavelength, and the distance from the origin representing the absorption at that wavelength. The group of spectra will have a structure in this 100-dimensional space. There will be greater variation in some directions than in others.

A new set of axis can be defined for the 100-one hundred dimensional graph, using the long axis of the group of spectra as the basis for the new set of axis. This gives an axis along which the greatest variation in spectra occurs. If this process is repeated, a second axis is obtained along which the second greatest variation occurs, then a third, and a fourth, and so on. Even though there are 100 wavelengths, the majority of the variation, and thus the bulk of the information available in the data, is described by a small number of these principal components. Note that these principle components are not necessarily specific wavelengths; they are chosen to express the greatest variation in the data; therefore, they could fall in any direction in the original 100-dimensional space. Fig. 11.3-4 shows the results of a PCA, in this case, three components. The upper plot (A) shows the first two principal components of the analysis. Each vertical group of points represents the spectra of a

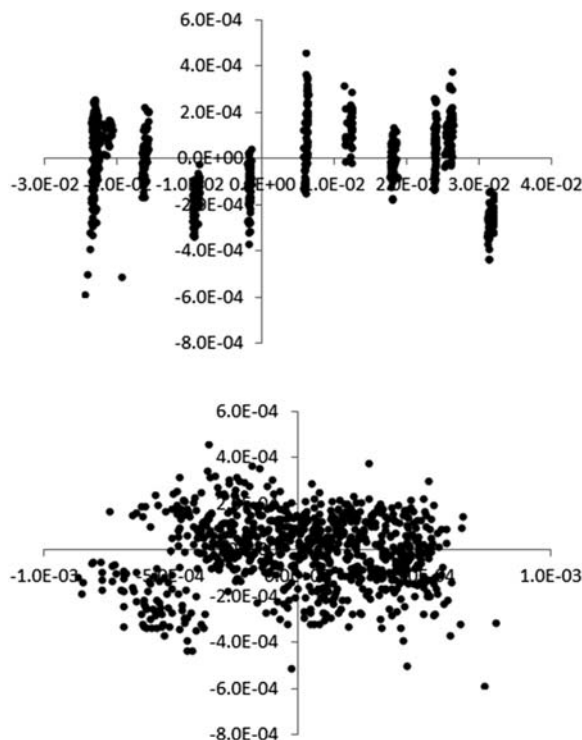


FIGURE 11.3-4 Plots showing PCA results for a three-component system. (A) The correlation between the first two principal components of the analysis. (B) The lack of correlation between the second and third components. *Used with permission of the authors.*

specific sample. The cloud of points in the lower graph (B) shows the second and third components of this analysis. In [Fig. 11.3-4B](#) the differences between the samples are no longer visible, because most of that difference is expressed in the first two principal components, [Fig. 11.3-4A](#).

[Fig. 11.3-5](#) shows the NIR spectra of five different complex chemical mixture, with several components, such as those analyzed in [Fig. 11.3-4](#). The X-axis is wavelength, while the Y-axis is the value of the model at that wavelength. When using NIR [\[36\]](#) to measure the concentration of chemicals, the measured spectrum is multiplied by the calibration model, to give the chemical concentration. At 1 and 3 μm there is a clear linear correlation between light absorption and concentration. The absorbance changes as a linear function of the concentration, but at about 2.25 μm a change in absorbance is noted; however, there is no linear correlation.

Chemometrics, implemented by using the PLS method, determines the optimal calibration model, which are the weights that will give the best estimate for the concentration of the component of interest. In the example shown in [Figs. 11.3-4 and 11.3-5](#), the calibration model will have a correlation value

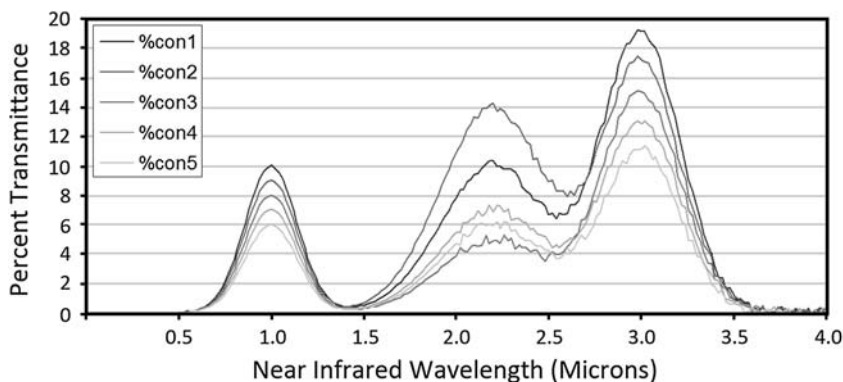


FIGURE 11.3-5 Near-infrared spectra of a complex chemical mixture, with several components, are shown for five different concentrations (%con) of one component. *Used with permission of the authors.*

of 0 at 2.25 μm , meaning that this part of the spectrum does not correlate well with the concentration. The spectrum at 1 μm would receive a larger correlation value than the spectrum at 3 μm , because the spectrum at 1 μm contains better information about the concentration, in this case at 3 μm the signal is noisier than at 1 μm , as seen in the wavy nature of the absorbance peak.

11.3.5.2 Calibration Procedure

To create a chemometrics calibration model for a particular application, a set of measured data must be collected, with known chemical concentrations related to its known absorption wavelength; this is termed the reference data. These data are modeled using a partial least squares fit algorithm, to remove the errors, which generates the calibration model. The errors could arise from mixing the chemical components accurately, to analytical tool measurements, to tool sensitivity to temperature, discussed further in this section. When the calibration model is used to calculate the chemical concentration in a fluid, the calibration model results are simply multiplied by the spectrum, wavelength by wavelength. The sum of the resulting array, with an added offset, will give the concentration. Note that there is usually a constant offset value added to the concentration result. This is because at zero chemical concentration, the composition is usually 100 wt% H_2O , which has a significant absorption, so the calculation will not return 0 wt%, as the concentration of the additive. Fig. 11.3-6 shows a schematic diagram of the steps involved in the calibration procedure.

There are several points to consider when performing a calibration: (1) coverage of the complete range of relevant chemical compositions, (2) coverage of the complete range of temperatures, (3) use of laboratory reference data, (4) interpretation and understanding of the calibration results, and (5) independent verification of the results.

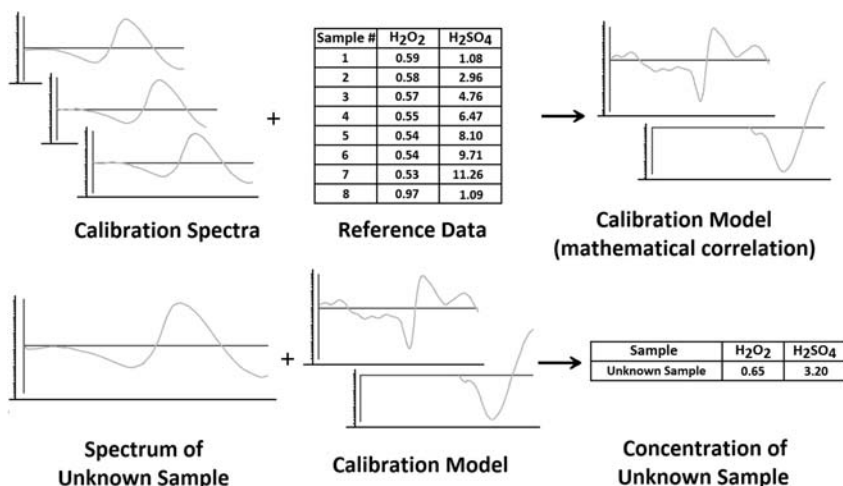


FIGURE 11.3-6 A schematic diagram for a chemometric calibration model showing the steps involved to determine the concentrations of a two-component mixture. *Used with permission of the authors.*

In addition, there are several points to consider when preparing a set of samples for calibration to ensure the model obtained is repeatedly applied to subsequent analysis.

The **accuracy** of the system will never be better than the accuracies of the reference concentrations, so care must be taken that these are accurately known. Chemical concentrations can be calculated from mixing ratios or obtained from laboratory analysis, such as gas chromatography or titration. When using mixed components in known proportions, the concentrations of the starting materials must be known and the quantities used must be measured as accurately as possible. When measuring concentrations with laboratory analysis, the accuracy of the method must be taken into account as well.

Independent methods of **chemical analysis** are called reference methods. Different reference methods will not necessarily exactly match each other. Titration or gas chromatography results may not be identical to each other or to the results obtained from the original mixing ratios of a sample. Even different laboratories using the same technique can differ somewhat from each other. It is best to use only one type of reference data in a calibration set.

The use of **reference methods** should be consistent when verifying a calibration sample. If the calibration sample used gas chromatography, then any verification work should be done using the same method, preferably in the same laboratory.

Some **chemicals are unstable**. The concentration of some materials will deteriorate over time, whether in storage or while being transported to a laboratory for analysis. In some cases the chemicals must be quickly cooled, then kept refrigerated until analysis.

It is not always possible to prepare a *complete set of calibration samples*. For example, if significant levels of residues, in the form of precipitation or emulsion, are introduced into the solution during the process, these will affect the spectra and the calibration as well. These include processes that etch large amounts of silicon or photoresist strippers that can be loaded with large amounts of photoresist. In some cases a laboratory calibration can be fine-tuned using field data to create a more robust calibration.

The *Beer–Lambert law is correctly applied* to concentrations measured in molarity. This is proportional to the number of molecules per given volume, which is what the light “sees” when going through a sample. In the semiconductor industry, concentrations are often measured in units of weight percent, which are not necessarily linearly related to molarity (mol L^{-1}), because they do not take volume into account. To clarify: although 1 kg of A mixed with 1 kg of B will result in 2 kg of a mixture, taking 1 L of A and mixing with 1 L of B does not necessarily result in 2 L of the mixture. The nonlinearity of weight percent units is generally not noticeable over a narrow range of concentrations, but over wide ranges it can be significant.

Temperature will affect the absorption spectra. Generally processes are run at a constant temperature, but when this is not the case, temperature variations must be addressed. Common methods include (1) taking a small sample stream from the process, cooling it, and stabilizing its temperature to a set value, (2) measuring the fluid temperature and using an algorithm to compensate for temperature changes, or (3) including a range of temperatures in the calibration set, to “teach” the calibration how to deal with temperature variations.

Some chemicals, notably, hydrofluoric acid (HF), behave differently depending on their concentrations. At very low concentrations, HF will dissociate to H^+ and F^- ions. At higher concentrations there will be more HF molecules, and a variety of other HF ions, such as H_2F_2 and H_2F^- [37]. The presence of other acids or buffering agents such as ammonium fluoride (NH_4F) can also affect the equilibrium between different ionic forms of HF.

11.3.6 Optical Absorption Instrumentation

There are several methods available for measuring the optical absorption spectrum of a chemical. For these methods there is at least one light source, some means of separating the different wavelengths of light, and a detector. Light sources include halogen lamps that produce light over a wide range of wavelengths, but require periodic replacement, and light-emitting diodes (LED) that have a long lifetime, but have a limited wavelength range.

11.3.6.1 Nondispersive Absorption

Dispersive optical absorption instruments separate the light, whether visible, UV, or near IR, into a spectrum of wavelengths that compose the light source

[38]. In a nondispersive instrument, the desired wavelengths of light are selected or rejected, usually by means of optical filters, from the light source [39]. These filters are typically discrete waveband filters, mounted in a rotating filter wheel. The filters must be chosen so that the relevant wavelengths of light can be measured for the target application, as each application requires a specific set of filters. Only one wavelength is measured at a time, which limits the light throughput of the system. These instruments tend to be relatively simple and inexpensive [36].

A continuous variable filter, in which the selected wavelength of light changes continuously over the diameter of the wheel, can also be used. These are available for a variety of wavelength ranges. However, instruments based on this technology are not being used in the semiconductor industry.

An alternate design, in which LEDs serve as wavelength specific elements, results in a simple, reliable instrument, although one that is limited in application, as are most nondispersive devices.

11.3.6.2 Dispersive Absorption

The most common spectrometers use diffraction gratings to split the light into its wavelengths. These gratings have a series of parallel lines inscribed in them that split light into wavelengths through an interference effect. They are usually used in a reflection mode, where the grating has a reflective coating.

In a scanning spectrometer, there is a single detector, and the grating is slowly rotated to cause different wavelengths of light to hit the detector [40], as shown in Fig. 11.3-7. These systems are relatively slow, include mechanical parts, which can be unreliable, and “throw away” most of the light signal, as it observes only one wavelength at a time, whereas the rest of the light misses the

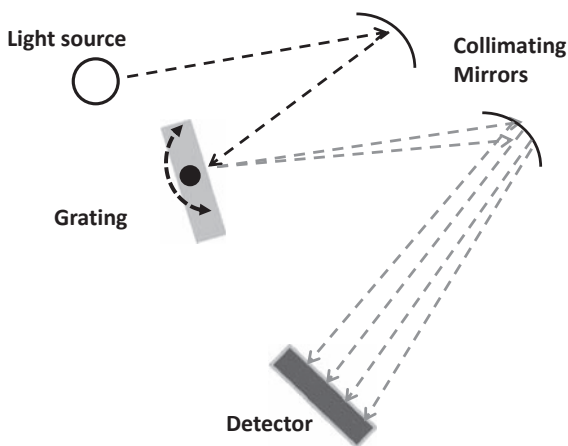


FIGURE 11.3-7 A schematic of a scanning grating spectrometer [40]. *Used with permission of the authors.*

detector. This is an archaic design that is rarely used today. In a photodiode detector array (PDA), the light that is split into separate wavelengths strikes the array, so that different wavelengths of light strike different individual detectors [36]. The advantage is that a complete spectrum can be recorded very quickly because it is detected all at once. There are no moving parts and there is more light throughput, because no light is thrown away. Fig. 11.3-8 shows a schematic of a PDA.

Each spectrometer covers a certain wavelength range. In a diode array instrument the light is spread over the detector plane, so a group of wavelengths hits each detector element. This is one of several factors that affect the resolution of a spectrometer, its ability to resolve between different wavelengths of light. The resolution of a spectrometer is usually given in nanometers as FWHM (full width at half maximum). This describes the width of the detected peak when using a single-wavelength light source.

Spectrometer stability over time is an important issue. There are a number of factors that can cause instrument drift. Instrument manufacturers take these into account and provide for correction of these sources of drift.

Light source intensity can change over time. This can be compensated for by measuring the light source intensity at each wavelength and using those measurements as reference values to compensate for the change in light source properties.

Mechanical changes in the optical system can also cause drift over time. The most common causes of this are varying environmental temperature and mechanical shock. The major effect is a shift in the wavelength assignment in the spectrum. Instruments are designed to minimize these effects. In addition, manufacturers add automatic wavelength calibration features to their instruments that check the wavelength calibration against a known, stable source

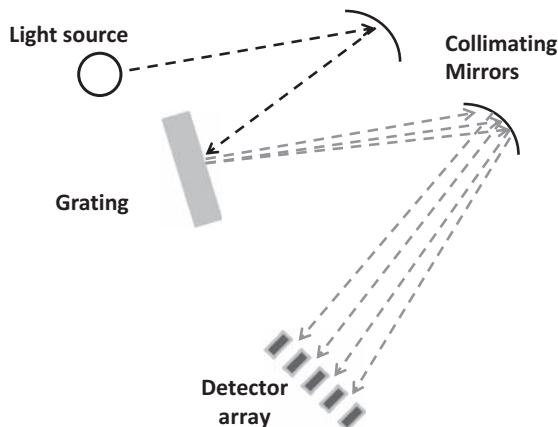


FIGURE 11.3-8 A schematic of a photodiode array [36]. Used with permission of the authors.

of wavelength information, such as a filter with known absorption peaks or a laser at a known wavelength.

11.3.6.3 Fourier Transform

Fourier transform is a technique used for optical interference to generate a complete spectrum using a single detector. Light is split into two beams. The optical path length that one of the beams travels is varied cyclically over time, usually with a moving mirror. The other beam goes through the sample and is modulated by the chemical makeup of the sample. Then the two beams are recombined before reaching the detector. The two light beams interfere, so that the light intensity at the detector varies over the optical path. This time-variant signal can be converted into a wavelength spectrum by taking the Fourier transform of the signal [41]. Fig. 11.3-9 depicts a schematic of a detector using the principle of Fourier transform.

The method has the simplicity of a single detector, with the advantage of relatively high light throughput. It does require fine mechanics to ensure proper motion of the moving mirror. Reading the detector output at high speed reduces the signal to noise ratio, although the ability to use a larger detector overcomes this to some degree [42]. For longer-wavelength IR, a single detector has a cost advantage over an array detector.

A downside of the Fourier transform method is that, for correct estimation of the spectrum, the absorbance of the fluid sample must be constant over the duration of the acquisition. Any passing bubble will result in a disturbed spectrum.

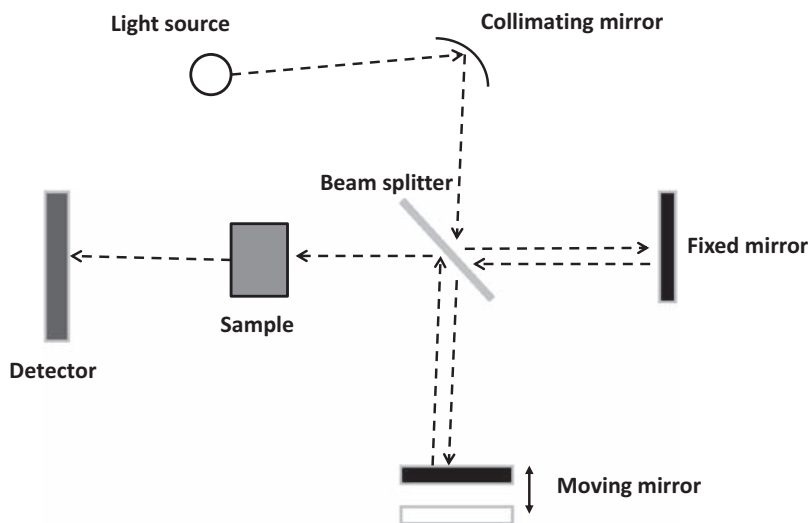


FIGURE 11.3-9 Schematic of a Fourier Transform Spectrometer.

11.3.7 Near-Infrared Instrumentation for Semiconductor Fabs

Chemical monitoring can be used to (1) obtain the formulation of a chemical mixture, (2) observe deterioration of the chemical or mixture over time, and (3) observe the formulation after spiking or replenishing the chemicals while processing wafers. A chemical monitor can be installed in a chemical mixing area (typically in the utility area), but more often in a recirculation line in the wafer processing tool itself. In single wafer wet processing tools the flow cell is typically installed on the supply tank recirculation line or at the point of use, whereas in batch tools it is installed on the bath recirculation line. There is interest in the semiconductor industry to measure closer to the point of use, but it is difficult to implement, as there are many delivery points, and the time response must be very short.

For the spectrometer to measure the fluid's absorbance of light, there must be some means of guiding the light through the fluid. A flow cell is such a means. An in-line flow cell is composed of a chemical-compatible tube where the chemical can flow through the light beam, discussed later in this section. Note that there is no chemical contact with the measurement system, only with the cell. The cell is typically installed in the recirculation line in a batch wafer wet processing tank or at the point of use for a single wafer wet processing module. This can be performed by the tool manufacturer when building a new tool or can be retrofitted onto an existing tool. For the retrofit, the work usually must wait for periodic maintenance when the tool can be shut down and rinsed before cutting the tubing lines, for safety reasons. Multiple flow cells can be used on one tool, whether for multiples of the same chemical baths on a batch tool or for different chemistries. Each bath will have a dedicated flow cell, with a common, multiplexed, monitoring instrument.

An alternate method to in-line flow cells is to bring the sample fluid to the measurement instrument via a narrow diameter tube. This has the advantage of allowing sample conditioning by stabilizing the temperature and removing the bubbles, which will be discussed in the next section. The drawback of this approach is that the fluid is flowing to the instrument, sometimes outside of the tool, which can be a safety issue. It can also result in a significant time delay in the measurements, depending on the volume of the tube and the distance from the wet bench tank.

Finally, there are “clip-on” in-line flow cells that are easy to install on an existing or new recirculation line. There is no need to cut the line because the absorption spectrum is measured directly through a Teflon-based tubing of the chemical recirculating line. The advantage is ease of installation. However, owing to the changing mechanical and optical properties of the tubing over time, affecting of the angle of incidence of the light drifts in signal can be observed [42].

In-line flow cells are usually connected to the measurement instrument with fiber optic cables, suitable to the wavelength of light. Note that these must

be handled with care, as sharp bends can crack the fibers. Multiple fiber optic cables run from the measurement instrument to the point of measurement. The measurement instrument is typically located on board and interfaced to the software of the wet processing tool in a closed loop control system.

Flow cells for NIR monitoring are usually made with materials that are standard to the semiconductor industry. Flow cell bodies have wetted parts of Teflon (PTFE or PFA), Kalrez, or similar types of materials seals, plus sapphire windows. Sapphire has good chemical resistance and does not leach ions into the fluid. In some cases windows are made of fused silica, which has excellent optical properties and is resistant to most chemicals, with the important exception of HF and other SiO_2 (silicon dioxide) etching agents. The flow cells must block out light originating from the ambient environment.

11.3.7.1 Factors That Interfere With Measurements

There are a few fluid properties that can interfere with measurement in the NIR. Bubbles are the most common issue. Bubbles are usually a result of H_2O_2 decomposition, in mixtures such as SC-1 or SPM. There are several ways of dealing with this. Careful plumbing arrangement can reduce the amount of bubbles that reach the flow cell. Bubble removers, also called bubble cutters, are an ancillary piece of equipment added to the recirculation line, which separate the flow into two streams, one of which contains most of the bubbles, whereas the other is mostly bubble-free. They make use of gravitational or centrifugal forces to separate the bubbles from the fluid [43].

Note that the bubble removers do not separate the smallest bubbles from the fluid. Also, the fluid is constantly generating new bubbles, which may reach the flow cell. Some in-line NIR instruments have smart algorithms that can deal with these bubbles. They sense when the optical signal is low, due to a bubble, and ignore the readings that have been affected by a bubble.

Instruments that have the fluid flow in a small tube can cool the fluid before removing the bubbles. This is an advantage, as fewer bubbles are generated once the fluid has cooled down. However, the cooling time contributes to a slower response time.

11.3.7.2 Response Time

The response time, defined as the time that passes before a change in the reading is reported by the NIR instrument, is usually in the range of a few seconds. The sample time interval is related to the ability of the monitoring system to measure and record the data obtained from the process tool. The minimum response time is related to the sampling time interval, but there are a number of other factors that can increase the response time: (1) fluid flow issues, (2) signal averaging, and (3) multiplexing.

The **fluid flow** to the measurement cell can be slow because of bubble removers, slow flow in a bypass line, or flow through a long, small-diameter

tube to the instrument. In these cases the NIR instrument may provide updated readings at a very high rate, but the readings will not reflect the current situation in the chemical tank, owing to the time delay in the fluid's arrival.

If the signal is noisy, due to interferences, such as bubbles in the flow, or due to the calibration model being very sensitive to noise in the spectrum, it may be necessary to apply *signal averaging* [44] to many readings before calculating an updated value for the concentration. This can lengthen the response time.

To reduce the cost of ownership of an in-line monitor, some instruments are able to measure several flow cells through a *multiplexing* arrangement. This is done by measuring each flow cell sequentially, one at a time. Measuring three cells takes about three times as long as measuring a single cell. Therefore, if time response is critical for an application, it is important to limit the number of flow cells being monitored by the same instrument.

11.3.7.3 Recalibration

NIR instruments may require repeat calibration for a number of reasons: light source replacement, changes in the optical system or, in some cases, periodic factory refurbishment and recalibration. On-site recalibration is generally a one-point calibration, or offset. This can be performed using a fluid with a known reference value. The most common methods are to use the concentration values of a known fresh chemical mix or DIW as a reference solution.

11.3.8 Example—Photoresist Stripper

In this example an NIR instrument was calibrated for a photoresist stripper chemical formulation. The calibration work was done in cooperation with Versum Materials (former Air Products and Chemicals), a manufacturer of a commercially available chemical. Note that the details of this case have been modified because of the proprietary nature of the information.

11.3.8.1 Design of Experiment

It is necessary to cover a wide range of concentrations when calibrating an NIR instrument. Temperature can also affect the concentration determination accuracy. The design of experiment (DOE) should take these factors into account. To cover all possible combinations of chemical composition, there must be a set of samples in which each of the chemical concentrations varies independently of the others.

For example, if only one chemical, such as NH_4OH , is dissolved in water, then a minimum of two samples at different concentrations would be needed, as the set is one dimensional (the H_2O content here does not vary independent of the NH_3 concentration). If there are two chemicals, for example, NH_4OH and H_2O_2 , then the data must be two-dimensional, and the concentrations will

lie on a plane in the concentration space. A minimum of three points will describe such a plane, although more points typically would be used in such a situation to improve the modeling accuracy. Three independent components will require samples that define a three-dimensional space, and so on.

In this example, there are six components, of which five are independent, whereas the concentration of the sixth component, H₂O, is 100 wt% minus the sum of the other components. This means that the sample set must span a five-dimensional space. The boundaries of this space are defined by the concentration range that each component can have. Table 11.5-2 shows the expected composition ranges that may be encountered.

If the components vary independently, then plotting various two-dimensional slices through this five-dimensional space, then checking if the sample concentrations are spread out properly over the space, this independence can be confirmed. This procedure may require many plots if there are

TABLE 11.5-2 Example of Experiment Design Component Mixture Ranges						
Process Information						
Component	A %	B	C	D	E	
Min	8	19	16	6	13	
Max	15	28	23	11	18	
Sample#	A	B	C	D	E	Water
1	10.56	22.71	19.06	7.95	14.92	24.80
2	10.70	23.02	19.30	8.04	15.11	23.84
3	10.57	22.76	19.52	8.09	15.10	23.96
4	8.48	19.59	22.79	8.12	15.13	25.90
5	14.37	22.05	16.43	6.13	13.83	27.18
6	10.50	19.62	22.71	5.96	15.11	26.10
7	8.42	26.37	16.40	10.16	15.26	23.40
8	8.32	23.83	15.93	10.47	14.25	27.19
9	8.33	19.26	20.21	10.65	14.15	27.41
10	12.56	19.15	16.25	10.69	12.80	28.54
11	9.41	26.85	22.46	6.23	17.44	17.61
12	12.35	27.51	16.41	5.96	15.64	22.12
13	14.45	20.14	19.84	6.02	14.33	25.22

many independent components. Fig. 11.3-10 shows a plot of expected concentration distribution for two of the six components.

Temperature is another concern with respect to accurate calibration. Although processes are usually run at a constant temperature, temperature variations do occur, and a fab may decide to change their process temperature, especially when the process is relatively new. Therefore, the DOE should include calibration samples that are measured over a range of temperatures. In such a case, the choice of the temperatures adds an extra dimension to the learning set.

For this study, the chemical manufacturer mixed the six components in varying proportions, per the planned experimental design. As the fluid is circulated at an elevated temperature, there is a concern that the more volatile components may evaporate and affect the composition of the mixture. To verify that this does not happen, chemical circulation is performed in a closed-loop system, and samples are analyzed both before and after the heating and circulation cycle. Small changes in concentration are observed, which are taken into account when associating concentrations with their spectra. After the experiment is completed, the data are analyzed using the PLS fit method to generate the calibration.

The calibration results for component A can be seen in Fig. 11.3-11. The values in the X-axis, “Measured Y” in the graph, are the concentration reference values. The values in the Y-axis, “Predicted Y” in the graph are the values calculated by the calibration model from the spectra. For a perfect calibration the points would all directly fall on a straight line, and the line would have a slope of exactly 1. In reality, there is always some variation from the ideal, because of the limited accuracy of both concentration reference data

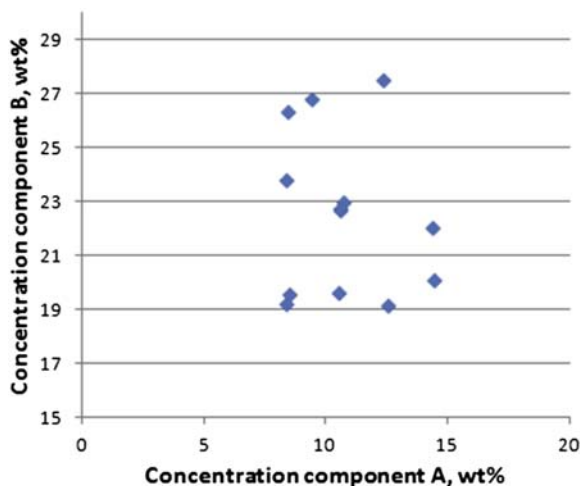


FIGURE 11.3-10 Plot of expected concentration distribution for two of the components of the six-component mixture. *Used with permission of the authors.*

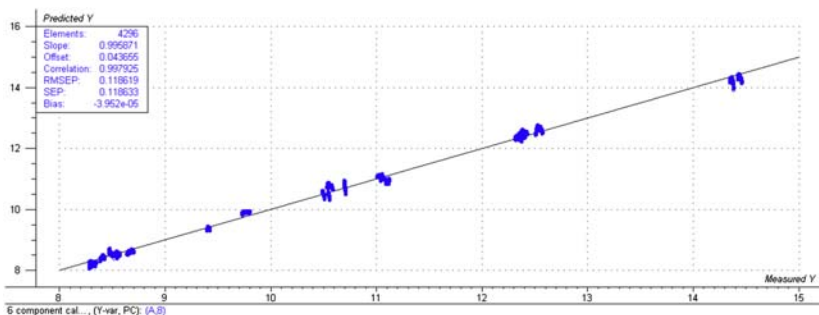


FIGURE 11.3-11 Calibration results for component A of the six-component mixture. *Used with permission of the authors.*

and the noise in the spectroscopic system. In this case, it is 0.996, which is quite good.

The calibration results also include some statistical information. The RMSEP, or root mean squared error of prediction, gives a measure of the expected RMS error of the calibration. The RMSEP is calculated by means of cross-validation. Using the cross-validation method, one spectrum at a time is removed from the set of spectra. A calibration model is then calculated using the other spectra, and the concentration value of the withheld spectrum is calculated (“predicted”) using the model. The error between the real value and the predicted value is calculated. The process is repeated for each spectrum. The square root of the average of the squares of these errors is the RMSEP. Note that the RMSEP gives a measure of the calibration’s accuracy; however, the optimal method to evaluate the calibration accuracy is by testing it with an independent validation set of samples, i.e., reference samples that were not included in the training set. This can be done by measuring a set of separate validation samples or, preferably, by taking samples from the process and having them chemically analyzed.

11.3.8.2 Photoresist Stripper—In-Line Monitoring

The calibration models were installed in a commercially available NIR analyzer in a fab. The instrument follows a commercially available aluminum postetch photoresist stripper formulation over time and helps determine when the chemical should be replaced, due to decomposition of the components. To field-validate the calibrations, samples were taken from the process at the same time of the NIR sampling and sent to the chemical manufacturer for analysis. Fig. 11.3-12 shows results of the change in concentration of a six-component mixture, one of which is water, over time. The graphic shows the change in the

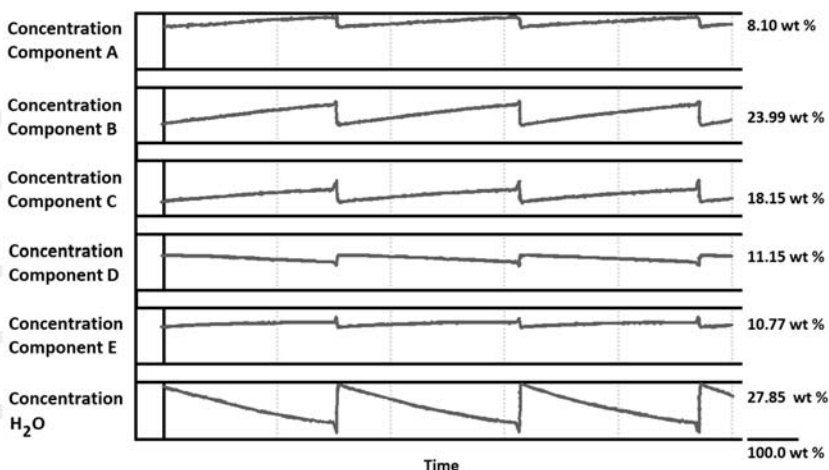


FIGURE 11.3-12 Graphical representation of the change in concentration of a six-component mixture, one of which is water, over time. *Used with permission of the authors.*

concentration and the recent history. Note the changes in the concentrations when dosing of the chemical mixture occurs, indicated by the spikes in concentration, for each component. The graphic may also indicate when a process is outside of its normal range.

11.4 ELECTROCHEMICAL METHODS: CONDUCTIVITY, RESISTIVITY, AND ION-SELECTIVE ELECTRODES

Electrochemical systems are traditionally defined as reactions that occur between an electrode and an electrolyte. The measurement of conductivity and resistivity uses electrochemical systems [45]. The principle of electrochemical systems involves charge transfer across an interface, which is the basis of the technical discussions in this section.

Both conductivity and resistivity are used to measure the purity of water and the concentration of chemicals in an aqueous solution. In addition, pH is also measured by using selective ion electrodes that measure conductivity of the H^+ ion. These monitoring methods will be discussed in this section.

11.4.1 Conductivity

Conductivity is a measure of the conductance of an electrical current through a material. Its unit is typically expressed in microSiemens/cm ($\mu S/cm$). In the case of wet processing chemicals, the material is an aqueous solution. For a current to be conducted through the aqueous solution, it must contain ions. The more the number of ions, also called electrolytes, the higher the ionic strength. For example, UPW has very low conductivity, because it has very few ions

available to conduct the electrical current, whereas aqueous HCl has high conductivity, because of the presence of many ions. Although the water molecule (H_2O) is not an electrolyte, it does have some conductivity, as hydrogen (H^+) and hydroxyl (OH^-) ions are present, due to the dissociation of water molecules. In addition, salts such as sodium chloride (NaCl) are electrolytes. Examples are shown:



Note that conductivity measures the concentration of all ions in solution because it cannot distinguish one ion from another.

11.4.2 Resistivity

Resistivity is the reciprocal of conductivity. Resistivity is a measure of the resistance of a material for a given length. Its unit is Ω cm. In applications such as UPW it is often used because the UPW ionic concentration, and thus the conductivity, is extremely low, for example, less than $1 \mu\text{S}/\text{cm}$; thus the resistivity scale is more convenient to use. The conductivity, κ , is defined as the reciprocal of the resistivity, ρ :

$$\kappa = \frac{1}{\rho} \quad (11.4-10)$$

11.4.3 Conductivity and Resistivity Measurements

There are various methods of measuring conductivity in aqueous solutions [46], including amperometric and potentiometric electrodes [47].

Amperometric probes were the first type of electrode developed for measuring conductivity. The electrodes are spaced 1 cm apart. Current is measured across the electrodes at a single applied voltage potential, V . The higher the current, I , the greater the conductance, C , according to Ohm's law:

$$I = \frac{V}{R} = V \times C \quad (11.4-11)$$

The ions in the aqueous solution migrate to the electrodes, the anions to the positively charged cathode and the cations to the negatively charged anode. However, salt deposition on the electrodes due to electrolysis can alter the resistance; therefore alternating current (AC) is used to prevent the deposition. This occurs because the alternating voltage, caused by the AC, causes the

capacitor to charge and then discharge, preventing deposition of electrolysis products on the electrodes. **Potentiometric** conductivity measurements are based on induction; these measurements typically use four concentric cylinders as electrodes, two outer rings that apply alternating voltage to induce a current in the solution and two inner rings to measure the voltage drop. An AC current is induced from the outer to the inner cylinder, and the voltage drop is measured between the two cylinders in the middle. This method eliminates the effects of polarization because, with alternate current, the polarization is limited to the volume near the outer electrodes and does not affect the reading. Typically, a shield is placed around the rings that helps maintain a constant electric field by maintaining a constant volume of solution with the rings, which will cause a nonlinear drop in voltage. Therefore, the four-electrode configuration can measure a wider range and higher conductivity solutions with more accuracy than the two-electrode configuration. The preferred metal for these electrodes is platinum because it can withstand high temperatures and corrosive chemicals and produces a stable reading.

Advanced conductivity instruments can vary the voltage applied to the electrodes, which enables the extension of the range of potentiometric measurements, and thus concentrations, even further. This allows advanced meters to be able to measure both high and low conductivities as well as the ultralow conductivity of UPW with one probe. The surface of the active electrode can incorporate a membrane that is ion selective to prevent ionic interference, creating an ion-selective electrode [48].

Inductive conductivity measurements have the advantage that the fluid does not wet the electrical parts of the sensor. For this measurement system, two coils are used. The first is the driving coil producing a magnetic field that is supplied with a known AC current. This coil forms a transformer together with the liquid, with the AC current passing through a channel in the sensor. The current induced in the fluid, in its turn, induces current in the second, pickup coil. The induced current is the output of the sensor. Typically, the two coils are integrated into one sensor head in the shape of a toroid.

Ion-selective electrodes (ISEs) [48–50], which include pH electrodes, selectively measure the potential of a target ionic species. These electrodes typically have a functional membrane that allows the species of interest to impart charge to the electrode, which allows a potential difference across the membrane. An ISE is made up of two electrodes, the first one is the electrode whose potential is determined by the solution to be tested and the second one is the reference electrode that is designed to have a constant potential. The difference in potential between the first electrode potential, E_{ise} , and the second reference electrode potential, E_{ref} , is the potential of the cell, E_{cell} :

$$E_{cell} = E_{ise} - E_{ref} \quad (11.4-12)$$

This in turn is proportional to the log of the concentration of the dissolved ion [47]. This relationship is described by the Nernst equation [51]:

$$E = E_o + \frac{RT}{zF} \ln \left(\frac{a_{ox}}{a_{red}} \right) \quad (11.4-13)$$

where E is the electrode potential (V), E_o is the electrode potential at standard state (1 M at 25°C and 1 atm), R is the ideal gas constant, T is the temperature, z is the number of electrons transferred, and F is the Faraday constant. a_{ox} and a_{red} are the activities of the oxidized and reduced forms of the ion, which are directly related to the concentration of the species in solution. Thus, the electrode potential is dependent on the concentration of the ion in the solution. The selective membrane is designed to ensure that only the ion of interest affects the cell potential. However, one should be aware that other ions can interfere with the measurement. For example, high concentrations of sodium ions can interfere with a pH electrode response. The selective membrane consists of selective ion-conducting materials; for example, for a pH electrode, the membrane is a special formula of glass. Other ISE membranes include crystalline or solid-state membranes, for example, LaF₃-doped crystal for fluoride selective electrodes, membranes composed of a special organic polymer that contains ionophores incorporated into an inert matrix, for Ca²⁺ and K⁺ monitoring, and gas-sensitive electrodes in which the gas permeates the membranes and reacts with the inner electrode solution, such as those used to detect NH₃ or CO₂ gas.

Ion-selective electrodes are attractive to use for monitoring ionic aqueous solutions due to their long lifetime, adequate ion selectivity, accuracy, and the wide concentration range that they are capable of measuring. The response time to reach equilibrium while the ions charge the membrane can be on the order of approximately 15 s [52], which is adequate for semiconductor monitoring. Note that ion-selective electrodes require a carefully controlled environment to operate correctly: constant temperature, constant pH (except for pH electrodes), and constant total ionic strength. Thus, a sample must be taken from the process stream and conditioned before the measurement can take place.

The **pH electrode** is based on potentiometric measurements with an ion-selective electrode, in this case made of thin doped glass with extremely high selectivity for hydrogen ions in aqueous solutions. Because HF can etch the glass of the pH electrode, other materials are available. Typically, the reference electrode is incorporated in the same probe as the potential electrode.

11.4.3.1 Effect of Temperature on Electrode Measurements

The Nernst equation (Eq 11.4-13) shows that the potential is dependent on temperature; thus a small temperature change, such as 1°C, can have an effect, typically 2%, on the electrode potential [53]. Thus, for semiconductor

applications, where processing temperatures can deviate from the standard 25°C, a compensation factor must be included to obtain the correct value for the concentration. Furthermore, the pH, resistivity, and conductivity of an aqueous solution change with temperature, because of the effect of temperature on the dissociation of weak acids and bases and the dissociation of water [54]. Light has studied the effects of temperature on water resistivity, from 0 to 100°C [55]. Ionic strength, dissolved CO₂, and interference from other ions in the solution will affect the measured value. These phenomena can explain discrepancies with in-line pH, resistivity, and conductivity measurements.

State-of the-art electrodes incorporate temperature compensation algorithms into the electrode's control software, with the realization that if the composition of the process changes, the temperature compensation coefficient must also change.

11.4.3.2 Calibration of Electrode Measurements

Calibration of electrodes, whether conductivity, resistivity, or pH, must be performed regularly.

Calibration of conductivity electrodes is performed to determine the effective cell constant. The cell constant, K_{cell} , is dependent on the length between the two electrodes, l , and the surface area of the electrodes, A , which is assumed to be identical for each of the electrodes and is dependent on the electrode material's surface roughness:

$$K_{cell} = \frac{l}{A} \quad (11.4-14)$$

The conductance value of a solution depends on that ion's mobility [56]. The mobility, μ , is dependent on the absolute value of the change of the ion, z_i :

$$\mu = \frac{z_i e}{6\pi\eta R_i} \quad (11.4-15)$$

where e is the electron charge, η is the viscosity of the solvent, and R_i is the ion's solvated radius, in this case, H₂O. The equation predicts that an ion, such as K⁺, that has a large R_i , will have a lower mobility than a smaller ion, such as H⁺, while a double charged ion, $z_i = 2$, will have a greater mobility than a singly charged ion, $z_i = 1$.

The conductivity electrode standard calibration solutions are dilute ultra-pure saline solutions, such as NaCl or KCl or a combination of these salts. Various conductivity solutions are available that comply with NIST standards [57]. Factors that affect the calibration standards are contamination and impurities in the standard solution, such as CO₂ that is dissolved in the aqueous solutions, forming additional ions from the H₂CO₃ [58], other inadvertent contamination from the environment, or improperly cleaning the electrodes. Contamination is especially deleterious when calibrating at the lower conductivity ranges.

Because conductivity probes measure only certain concentrations, the range of conductivity to be measured should coincide with the chosen conductivity calibration standard solution. Suppliers of conductivity probes suggest performing conductivity calibration with two different standards within the conductivity range of interest.

Calibration of resistivity electrodes uses the same procedures and similar NIST traceable solutions as conductivity probe calibration, with the chosen calibration standards appropriate to resistivity calibration.

Calibration of pH electrodes uses buffer solutions, usually at three standard pH values of 4, 7, and 10. The procedures are similar to the conductivity probe procedures.

11.4.3.3 Installation of Electrodes

The sensors in this section can be associated with handheld monitors or installed in a wet processing tool to monitor the chemical concentration for incoming chemicals, at the point of use or in-line during processing. For wet benches, the sensor must be installed so that the working part of the electrodes is completely immersed in the process liquid. Bubbles should be avoided; therefore installing the sensor in areas where air bubbles are likely to get trapped is detrimental to accurate readings. Avoid placing the sensor in an area that may experience dead space (air gap) instead of liquid. Do not install the sensor in dead legs or areas where circulation is poor. Also, precipitation that can accumulate on the electrodes should be avoided.

Mounting the sensor in a vertical pipe with the fluid flowing in an upward direction is suggested. If the sensor must be installed in a horizontal pipe, place the sensor at an angle of 180 degrees to the pipe. Keep adequate clearance between the end of the sensor and the opposite pipe wall to keep response time as fast as possible.

Instrumentation to control the electrodes and to record the data is interfaced to the wet processing tool.

11.4.4 Ultrapure Water Resistivity Measurements

The purity of UPW used in semiconductor processing is of utmost importance [59]. Not only for semiconductor applications, but also for the water industry, resistivity is used to classify water types with respect to cleanliness and purity. The term “ultrapure water” is designated by its high purity with respect to organic and inorganic compounds, particles, and dissolved gases, as defined in SEMI Spec F63-1016 or ASTM Spec D5127-13. The ITRS (International Technology Roadmap for Semiconductors) also publishes guidelines for UPW [60]. The theoretical limit to resistivity of UPW is 18.18 M Ω at 25°C [61], which is regularly generated by today’s commercially available water purification systems; an example is shown in [62].

In an IC manufacturing facility, the UPW that is generated is tightly controlled with respect to resistivity [63]. Monitoring of the UPW occurs at various places in the UPW manufacturing process and delivery system, plus at the point of use of the water, for example, during rinsing the wafers after wet chemical processing. Resistivity can be measured real time.

Contamination of the UPW can occur within the delivery and recirculating systems and when the water is exposed to chemical during a rinsing cycle, which is the focus of the discussion in this section. A UPW rinse cycle follows the chemical treatment of the wafers; as the chemicals are removed from the wafer surface, they dissolve into the rinse water. During the rinsing process the UPW is monitored using resistivity [64,65]. Monitoring the resistivity and the time or the number of rinsing cycles, for example, the number of dump rinse cycles or the time for an overflow rinse tank, can assist in optimizing the rinsing process.

Programs to reduce the amount of UPW used and the waste water it generates have used resistivity to monitor the water after rinsing to reduce its usage by optimizing the processing time [65,66]. The resistivity monitor can be interfaced with the wet processing tool's software to rinse to a preset resistivity. In addition, after rinsing the contaminated UPW can be recycled back into the feed loop of the UPW generator, because most postrinse water is cleaner than city water as determined using its resistivity and pH values, although not all rinse water is recyclable, such as the water from the photoresist stripping process, because of the type of impurities [67].

The ability to detect trace amounts of ions in the water using resistivity is the key to controlling the undesirable contamination. Resistivity instruments must be able to detect impurities of only a few tenths of a part per billion (ppb), which equates to micrograms per liter [68]. If additional analysis of the water is needed, techniques such as ICP-MS (inductively coupled plasma-mass spectrometry) can be used [69]. This technique is outlined in Chapter 13. Trace element analysis can be used to determine the impurities that are present and to ascertain the source of the contamination.

Even small amounts of contamination in UPW can affect the resistivity. Fig. 11.4-13 shows the change in resistivity of UPW with the addition of trace contamination at the parts per trillions (ppt) level. Note the change in resistivity occurs at concentration >10 ppt, which equates to ng/L. The guidelines for ITRS UPW specifications [60] state that the metal impurities should be <10 ppt for general metal contamination, and <1 ppt for critical metal contamination, such as Al, Ca, Cr, Cu, Fe, K, Li, Mg, Na, Ni, Ti, and Zn, although the total impurity concentration for each metal is not stated.

In some cases, in the semiconductor industry, measurement of the resistivity of UPW is required at temperatures that deviate from 25°C, for example, high-temperature water rinsing of viscous chemicals such as SPM, discussed in Chapter 4. It is especially important to remove the sulfate residue from the wafer surface after SPM; therefore monitoring the resistivity will

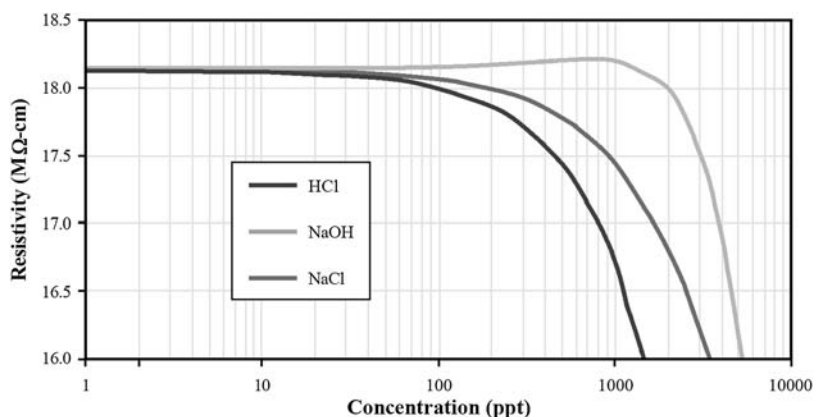


FIGURE 11.4-13 Graph of impurities; HCl, NaOH, and NaCl at ppt-levels in ultrapure water showing the change in resistivity. Only a small amount of impurities will decrease the resistivity. Adapted from P. Geladi, B.R. Kowalski, *Anal. Chim. Acta* 185 (1986) 1. by the authors.

assure that most of the ions are removed before the rinsing process is completed. As shown in Section 11.4.3, temperature affects the resistivity of UPW. It is more difficult to measure the purity of hot water than cold water [54,61,70] because of the temperature sensitivity, temperature measurement accuracy, and the need for calibration of the electrodes.

Calibration of resistivity electrodes is required because accurate measurement of the resistivity is key to determining the limit of detection for ionic impurities [64]. Without calibration, the relationship between the resistivity and the purity of water and the relationship to temperature is unknown. Calibration should be performed regularly, although it is seldom performed once the probe is installed onto the wet processing tool.

Temperature calibration is performed using a temperature standard or a low-conductivity solution, such as <10 mS/cm water at 0°C. The probe is immersed in the temperature standard or chilled solution and then the calibration is performed. Typically, the resistivity instrument has a standardized program for temperature calibration. Calibration of the conductivity is performed by using a standard solution of 1 M KCl (potassium chloride) [71]. ASTM standard D1125-14 outlines current methods for standardization of test methods for measuring the resistivity of water. Standards can be prepared using ASTM D1125-14 or NIST (National Institute of Standards and Technology) solutions [72].

11.4.5 Chemical Conductivity Measurements

Control of chemical concentration is critical for controlled and cost-effective processing in an IC manufacturing facility. Besides monitoring chemicals

with optical techniques, such as NIR that is discussed in [Section 11.3](#), conductivity provides rapid, cost-effective, and real-time control of chemical concentrations [73]. The conductivity probes can be fashioned out of chemical-resistant materials for use with corrosive chemicals, such as HCl and HF, or noncontact induction probes may be used. Conductivity monitoring is used to keep the wet cleaning and etching process, which involve chemicals that are electrolytes, under control over the lifetime of the bath. In addition, bath life, in particular when the bath should be replaced because the process can no longer be kept in control, can be determined by monitoring the conductivity.

Kashkoush et al. have used conductivity to monitor the concentration of HF in a SiO₂ etching bath [73]. To obtain a constant etch rate, the influence of the by-products of the etching process must be taken into consideration. The researchers assumed that they would observe a decrease in conductivity as the HF is consumed but, in fact, observed an increase in conductivity due to the by-products. The concentration of the HF is controlled by the periodic spiking of HF and the replenishing of H₂O into the bath, as determined by an algorithm that uses conductivity as the control parameter. Another contribution to the variation in conductivity is the temperature; the authors found that by holding the temperature constant and maintaining the conductivity of the HF solution, the etch rate of SiO₂ can be controlled.

11.5 FUTURE OF IN-LINE CHEMICAL MONITORING

Wet chemistries are becoming more and more dilute over time. In the past, an SC-1 monitoring system could deal with >3 wt% H₂O₂, and accuracies of >0.1 wt% would suffice. The trend today is toward solutions with the H₂O₂ concentrations in the range of a few tenths of a percent, and accuracies need to be in the hundredths of a percent. Monitoring these dilute solutions requires sensors to be both very sensitive and highly stable over time. Current efforts center on new designs of light sources and improved light reference measurements.

An instrument's intrinsic long-term stability can be improved by using zero-point calibration to compensate for long-term drift. This type of calibration requires the test solution to be replaced with a material with known spectroscopic properties. This is usually DIW, although in principle other reference solutions or solids can be used.

There are basic limitations to NIR measurements and optical methods in general, and alternate methods may have to be developed to deal with extremely dilute chemistries. In addition, new chemistries are needed as semiconductor processes advance to smaller and smaller nodes. Measurement methods need to be developed for these chemistries as the technologies rapidly change, and instruments must be flexible to handle these changing needs.

There are opposing needs in the field of chemical monitoring. On the one hand, repeatable, accurate, timely results are wanted, on the other hand, monitoring solutions need to be cost-effective, not just the purchase cost, but also the overall cost of ownership. Practical solutions must compromise between these conflicting requirements.

For example, online titration was historically used in fabs. It can provide accurate and repeatable results but has a relatively slow time response and a high cost of ownership, in terms of supplying various calibration fluids and the need to periodically replace electrodes. Over time online titration has been largely replaced by spectroscopic methods such as NIR absorption spectroscopy. NIR absorption spectroscopy provides a balance between performance and cost of ownership that is appropriate for most fab applications.

Today the trend is to more dilute solutions, and standard spectroscopic methods are reaching their limits. In addition, the acceptable concentration limits in new chemistries are quite narrow, making it risky to run process without monitoring the chemistry. This will lead the industry toward alternate methods, including wet chemistry methods reminiscent of online titrations, with some of the problematic aspects of such methods.

An example of this is DSP+. The sulfuric acid and hydrogen peroxide in this solution are at a high enough concentration that NIR methods can readily measure them. The critical component, however, is HF, which is present in the range of about 100 ppm [74]. This is beyond the capabilities of spectroscopic methods. Typically, F^- is measured using an ion-selective electrode. This in turn requires a number of reagents to be fed into the instrument: buffer solutions, ionic strength adjusting solutions, and calibration solutions. The electrode itself must be periodically replaced. This type of device is reminiscent of online titrators, with all of their inherent disadvantages; however, the device does have the potential to successfully measure the low levels of HF in the DSP+ solution.

In the future, there will likely be more applications that require complex and expensive solutions. This will be justified in terms of improved process control.

Another direction that we may see is the combination of existing methods, such as NIR, conductivity, refractive index, or the speed of sound to achieve better performance.

In-line measurement of metal contamination can monitor the bath until failure, for example, if a particular dirty wafer or the metal content in the chemical exceeds the specification limits, the process engineers will be alerted to the excursion.

11.6 SUMMARY

Process monitoring is essential in many semiconductor processes. As the industry moves to smaller nodes, processes, including wet processes, need to be

kept within tighter control limits, and the importance of continuous chemical concentration monitoring increases. In addition, new chemistries are being developed for these nodes, and instruments must be calibrated to measure them. NIR technology must keep pace with these changing needs. NIR spectroscopy, along with chemometrics, is a proven, flexible, and reliable method to monitor processes. Although NIR is rising to meet the demands of new applications, some processes require alternate, less convenient methods.

Measuring the resistivity of UPW has been and will continue to be an IC industry standard for measuring the quality of the water, whereas measuring the electrical conductivity of aqueous solutions can provide information about the concentration of the species dissolved in the solution.

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